

Hierarchical Clustering and Recommender Systems: Questions and Answers

Based on EECS 836 Machine Learning Notes

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Introduction

This document provides a comprehensive set of fundamental, computational, and conceptual questions and answers on hierarchical clustering and recommender systems, derived from the lecture notes of EECS 836: Machine Learning by Zijun Yao at the University of Kansas. Topics include hierarchical clustering algorithms (agglomerative and divisive), inter-cluster distance measures (MIN, MAX, Group Average, Ward's Method), computational complexity, and recommender systems (content-based and collaborative filtering, including matrix factorization).

1 Fundamental Questions

These questions focus on the basic concepts and definitions related to hierarchical clustering and recommender systems.

1. What is hierarchical clustering, and how is it visualized?

- *Answer:* Hierarchical clustering produces a set of nested clusters organized as a hierarchical tree. It is visualized as a dendrogram, a tree-like diagram where the x-axis represents data points and the y-axis represents the distance between clusters, recording the sequence of merges or splits.

2. What are the two main types of hierarchical clustering?

- *Answer:* The two main types are:
 - *Agglomerative:* Starts with individual points as clusters and merges the closest pairs until one cluster (or k clusters) remains.
 - *Divisive:* Starts with one all-inclusive cluster and splits clusters until each contains an individual point (or k clusters).

3. What is the role of the proximity matrix in hierarchical clustering?

- *Answer:* The proximity matrix stores the similarity or distance between pairs of clusters or points. It is computed initially and updated after each merge or split to reflect the distances between the new clusters and remaining clusters.

4. **What are the different methods to define inter-cluster distance in hierarchical clustering?**

- *Answer:* The methods include:
 - MIN (Single Linkage): Distance based on the closest pair of points between clusters.
 - MAX (Complete Linkage): Distance based on the farthest pair of points between clusters.
 - Group Average: Average pairwise distance between all points in the two clusters.
 - Distance Between Centroids: Distance between the centroids of the two clusters.
 - Ward's Method: Based on the increase in squared error when clusters are merged.

5. **What is the difference between content-based and collaborative filtering in recommender systems?**

- *Answer:* Content-based filtering recommends items based on their features (e.g., movie genres, actors) and user preferences, requiring side information. Collaborative filtering uses user-item interactions (e.g., ratings) to find similar users or items, not requiring side information but struggling with new items without ratings.

2 Computational Questions

These questions involve calculations, data analysis, or algorithmic procedures related to hierarchical clustering and recommender systems.

(a) **Using the MIN linkage distance matrix from the document (page 17), what is the first pair of points merged, and what is the distance?**

- *Answer:* The distance matrix is:

	1	2	3	4	5	6
1	0	0.24	0.22	0.37	0.34	0.23
2	0.24	0	0.15	0.20	0.14	0.25
3	0.22	0.15	0	0.15	0.28	0.11
4	0.37	0.20	0.15	0	0.29	0.22
5	0.34	0.14	0.28	0.29	0	0.39
6	0.23	0.25	0.11	0.22	0.39	0

The smallest distance is 0.11 between points 3 and 6. Thus, points 3 and 6 are merged first at a distance of 0.11.

(b) **For the Group Average method, calculate the proximity between two clusters $C_1 = \{p_1, p_2\}$ and $C_2 = \{p_3, p_4\}$ using the distance matrix from page 26, assuming points 1, 2, 3, 4 correspond to the first four rows/columns.**

- *Answer:* The Group Average proximity is:

$$\text{proximity}(C_1, C_2) = \frac{\sum_{p_i \in C_1, p_j \in C_2} \text{dist}(p_i, p_j)}{|C_1| \times |C_2|}$$

From the matrix:

$$\text{dist}(1, 3) = 0.22, \text{dist}(1, 4) = 0.37, \text{dist}(2, 3) = 0.15, \text{dist}(2, 4) = 0.20$$

Sum of distances:

$$0.22 + 0.37 + 0.15 + 0.20 = 0.94$$

Number of pairs: $|C_1| \times |C_2| = 2 \times 2 = 4$

$$\text{proximity}(C_1, C_2) = \frac{0.94}{4} = 0.235$$

Thus, the proximity is 0.235.

- (c) **Compute the Jaccard similarity between users A and B from the user similarity table on page 45.**

- *Answer:* The table is:

	HP1	HP2	HP3	SW1	SW2	SW3
A	1	0	0	1	1	0
B	1	1	0	0	0	0

Jaccard similarity:

$$\text{Jsim}(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

Intersection (both 1): Only HP1 (1 item). Union (at least one 1): HP1, HP2, SW1, SW2 (4 items).

$$\text{Jsim}(A, B) = \frac{1}{4} = 0.25$$

Thus, the Jaccard similarity is 0.25.

- (d) **For the collaborative filtering rating prediction on page 48, calculate the predicted rating for user u on item i using user-user collaborative filtering with two similar users v_1 and v_2 , where $\text{sim}(u, v_1) = 0.5$, $\text{sim}(u, v_2) = 0.3$, $r_{v_1, i} = 4$, and $r_{v_2, i} = 5$.**

- *Answer:* The formula is:

$$\bar{r}_{ui} = \frac{\sum_{v \in \text{Top-}K(u)} \text{sim}(u, v) r_{vi}}{\sum_{v \in \text{Top-}K(u)} \text{sim}(u, v)}$$

Numerator:

$$(0.5 \times 4) + (0.3 \times 5) = 2 + 1.5 = 3.5$$

Denominator:

$$\begin{aligned} 0.5 + 0.3 &= 0.8 \\ \bar{r}_{ui} &= \frac{3.5}{0.8} = 4.375 \end{aligned}$$

Thus, the predicted rating is 4.375.

- (e) **Compute the Root Mean Square Error (RMSE) for a recommender system with predicted ratings $\hat{r}_{ij} = \{4.5, 3.0, 5.0\}$ and actual ratings $r_{ij} = \{4.0, 3.5, 4.5\}$.**

- *Answer:* The RMSE formula is:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i,j} (\hat{r}_{ij} - r_{ij})^2}$$

Differences:

$$(4.5 - 4.0)^2 = 0.25, \quad (3.0 - 3.5)^2 = 0.25, \quad (5.0 - 4.5)^2 = 0.25$$

Sum:

$$0.25 + 0.25 + 0.25 = 0.75$$

$$\text{RMSE} = \sqrt{\frac{0.75}{3}} = \sqrt{0.25} = 0.5$$

Thus, the RMSE is 0.5.

3 Conceptual Questions

These questions explore the deeper understanding and implications of hierarchical clustering and recommender systems.

- i. **Why does hierarchical clustering not require specifying the number of clusters in advance?**

- *Answer:* Hierarchical clustering produces a dendrogram that represents a hierarchy of clusters. By cutting the dendrogram at different levels, any desired number of clusters can be obtained, eliminating the need to specify the number of clusters beforehand, unlike methods like K-means.

- ii. **What are the strengths and limitations of the MIN (Single Linkage) method in hierarchical clustering?**

- *Answer:*
 - *Strengths:* Can handle non-elliptical shapes, as it merges clusters based on the closest pair of points, allowing for elongated or irregularly shaped clusters.
 - *Limitations:* Sensitive to noise, as a single noisy point can cause incorrect merges, leading to poor clustering results.

- iii. **How does the MAX (Complete Linkage) method differ from MIN, and why is it less susceptible to noise?**

- *Answer:* MAX uses the farthest pair of points between clusters to determine proximity, unlike MIN, which uses the closest pair. This makes MAX less susceptible to noise because a single noisy point does not dominate the merge decision; instead, it considers the overall separation between clusters, but it tends to break large clusters and favors globular shapes.

- iv. **Why is Ward's Method considered a hierarchical analogue of K-means?**
- *Answer:* Ward's Method minimizes the increase in squared error (SSE) when merging clusters, similar to how K-means minimizes the within-cluster SSE. Both methods aim to form compact, globular clusters, and Ward's Method can be used to initialize K-means by providing a hierarchical structure to guide cluster assignments.
- v. **What are the computational challenges of hierarchical clustering, and how can they be mitigated?**
- *Answer:* Hierarchical clustering requires $O(N^2)$ space for the proximity matrix and $O(N^3)$ time due to repeated updates and searches of the matrix for N points. The time complexity can be reduced to $O(N^2 \log N)$ using efficient data structures (e.g., priority queues) or approximations, but the space requirement remains a challenge for large datasets.
- vi. **Why does content-based filtering struggle with new users, and how does collaborative filtering address this?**
- *Answer:* Content-based filtering requires a user profile built from item features, which is difficult for new users with no prior interactions. Collaborative filtering leverages ratings from similar users, allowing recommendations for new users by finding others with similar tastes, but it struggles with new items lacking ratings (the cold-start problem).
- vii. **What is the advantage of matrix factorization in collaborative filtering, and how does it handle sparse data?**
- *Answer:* Matrix factorization represents users and items as low-dimensional vectors, predicting ratings as their dot product. It handles sparse data by learning latent factors from observed ratings, generalizing to unseen user-item pairs, and regularizing to prevent overfitting, making it effective for large, sparse datasets like those in recommender systems.
- viii. **Why is the Netflix Prize significant in the context of recommender systems?**
- *Answer:* The Netflix Prize offered a \$1 million prize to improve prediction accuracy by 10%, spurring advancements in collaborative filtering, particularly matrix factorization techniques. It highlighted the importance of scalable, accurate recommender systems and fostered research in handling large-scale, sparse datasets.